Stable marriage (matching) problem

1. Show lists of preferences with at least 2 stable matchings (Try $n = 2$.)
2. Show a stable matching that is not the best stable matching for any participants of the matching.
3. Is it possible, that everyone gets their 2nd choice in a stable matching?
4. Is it possible, that everyone gets their 3rd choice in a stable matching?
5. Show an instance of SMP with n boys and n girls, such that the number of stable matchings is at least exponential.
6. Assume that A and B are a pair in a stable matching even though they are at the last position on their pair's list. Show, that they are a pair in every stable matching.
7. Assume that all girls have the same list of preferences. Prove that there exists at most one stable matching.
8. Stable roommate problem. Given $2n$ students and n rooms of capacity 2. Each student has a list of preferences for a roommate. Show an instance without a stable matching.